Transient Modeling of TSV-Wire Electromigration and Lifetime Analysis of Power Distribution Network for 3D ICs

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***Abstract*— In this paper, we present a transient modeling of electromigration (EM) in TSV and TSV-to-wire interfaces in the power delivery network (PDN) of 3D ICs. In particular, we model atomic depletion and accumulation, effective resistance degradation, and full chip-scale PDN lifetime degradation due to EM. Our major focuses are on: (1) time-dependent multi-physics EM modeling approach to model TSVs and connecting wires under the influence of coupled physical phenomenon including electric field, temperature, and stress; (2) time-dependent EM-aware power integrity analysis methodology, which is integrated with the TSV modeling approach to predict long-term IR-drop degradation in full-chip 3D power delivery networks. Our studies show that voids and hillocks grow at various TSV-to-wire interfaces and degrade the effective resistance of TSVs significantly. In addition, our full-chip PDN lifetime analysis shows significant increase in maximum IR drop during lifetime due to EM effects.**

I. Introduction

Through-silicon-via (TSV)-based 3D integration has recently received a lot of interest due to its potential to overcome conventional CMOS scaling limitations and its potential to enable heterogeneous integration. Reliability of TSV-based 3D ICs is an important issue for main stream acceptance. Electromigration (EM) can reduce IC reliability and potentially cause short or open circuit failures. TSVs, especially the power/ground (P/G) TSVs in 3D power delivery networks (PDNs), carry large currents and typically have higher local current densities due to current crowding [1]. Regions of high local current density are more susceptible to EM degradation. Therefore, long-term EM reliability issue for TSV-based 3D PDNs needs to be addressed.

EM modeling of interconnects has been extensively studied for many decades [2], [3]. However, very little work has been done to model the reliability of TSVs. A recent paper investigated DC current crowding inside TSVs and at the connections between TSVs and power wires [1]. Current crowding increases the effective resistance of the TSV and voltage drop in the 3D PDNs. The impact of TSV stress on back-end-of-line (BEOL) interconnects and its EM lifetime were modeled [4], [5]. Another modeling approach for TSV EM reliability was proposed in [6], where EM measurements on a TSV connection indicates that voids may occur at the TSV-to-wire interfaces [7]. However, none of these works discuss time-dependent EM modeling for TSV-based 3D connection or study long-term reliability of chip-scale PDNs.

Since atomic diffusion is dominated by the fast diffusion along grain boundaries compared to grains, any realistic EM modeling needs to include grain and grain boundary geometry. Most existing works on TSV EM modeling do not consider grain structure [4], [5], [6], but instead model the entire TSV as a single perfect crystal. The grain size distribution measured from actual TSVs can vary significantly, from 2.3um [8] in mean, 2.0-5.5um [9], or 1.0-10.0um [10] depending on the process technology.

The focus of this work is to model and investigate EM and reliability of TSV-based 3D connections and to analyze the impact on the lifetime of 3D PDNs. Our contributions are as follows: (1) multi-physics transient modeling is used to model and simulate EM in TSV-based 3D connections. This approach enables simultaneous consideration of the atomic transport in the structure, evolution of electric current, stress distribution, atomic concentration, and TSV effective resistance. (2) We investigate how different physical effects and parameters impact EM in TSVs and ultimately the PDN itself including the effects of temperature, electric current, back flow and thermal expansion stress, activation energy and grain structure. (3) We present a methodology for EM reliability analysis of full-chip-scale 3D PDNs. This methodology integrates our TSV EM modeling with time-dependent full-chip PDN analysis. Simulations show that the degradation of PDN IR-drop due to current crowding and EM can be modeled and quantified. Related experiments show that the maximum IR drop can increase 5% to 9% due to current crowding and later increase approximately by 11% to 26% during life time due to current and stress-driven EM effects.

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II. Transient EM Modeling

A. Fundamental EM PDEs

A set of well-known partial differential equations (PDEs) is used to model the time-dependent evolution of atomic concentration $N(x, y, z, t)$ at location (x, y, z) at time t.

$$\frac{\partial N}{\partial t} + \nabla \cdot \mathbf{J} = 0 \tag{1}$$

$$\mathbf{J} = \mathbf{J_c} + \mathbf{J_T} + \mathbf{J_s} + \mathbf{J_N} \tag{2}$$

$$\mathbf{J_N} = -D\ \nabla N \tag{3}$$

$$\mathbf{J_c} = \frac{N}{kT} e Z^* \rho \mathbf{j}\ D \tag{4}$$

$$\mathbf{J_T} = -\frac{NQ^*}{kT^2}\ D\ \nabla T \tag{5}$$

$$\mathbf{J_s} = \frac{N\Omega}{kT}\ D\ \nabla \sigma_{\mathrm{H}} \tag{6}$$

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TABLE I
NOTATIONS, MEANINGS, AND TYPICAL VALUE IN OUR EM PDES.

Term	Meaning	Typical value
N	Atomic concentration	N_0=1.53e28 Atoms/um^3
E_{A}	Activation energy	E_{A}(g)=2.1eV; E_{A}(gb)=0.8eV
k	Boltzmann constant	k=1.38e-23 J/K
e	Electric charge	e=1.6e-19C
Z^*	Effective valence charge	Z^*=-4[13]
ρ	Electrical resistivity	ρ_0^*=1.8e-8Ω·m
D_0	Self-diffusion coefficient	D_0=1e-7 m^2/s
Q^*	Heat of transport	Q^*=1.387e-20
Ω	Atomic volume	Ω=1.182e-29m^3
B	Back flow stress modular	5e7Pa
$\mathbf{j}$	Current density in mA/um^2	from simulation
T	Absolute temperature in K	from simulation
σ_{H}	Hydrostatic stress in Pa	from simulation
σ_{bf}	Back flow stress in Pa	from simulation

$$\sigma_{\mathrm{bf}} = B\frac{N - N_0}{N_0} \tag{7}$$

$$D = D_0 \exp(\frac{-E_{\mathrm{A}}}{kT}) \tag{8}$$

The continuity equation (Equation (1)) governs the atomic flux divergence ($\nabla \cdot \mathbf{J}$) over spatial dimensions and determines how the atomic concentration evolves over time. The atomic flux $\mathbf{J}$ is determined by the combined mechanisms described in Equations (3) through (6), where $\mathbf{J_N}$ is governed by atomic concentration gradient ∇N; $\mathbf{J_c}$ by current density $\mathbf{j}$; $\mathbf{J_T}$ by thermal gradient ∇T; and $\mathbf{J_s}$ by stress gradient $\nabla\sigma_{\mathrm{H}}$. The hydrostatic stress σ_{H} primarily involves the back flow stress σ_{bf} in Equation (7). The stress modeling and simulation will be discussed in detail in Sections II-C, IV-C, and IV-D. COMSOL multiphysics, a commercial finite element simulation tool [11], is used to build and solve these PDEs and conduct electric, thermal, and stress simulation. The notations are defined in Table I.

B. Grain and Grain Boundary Modeling

The generic TSV model used in this study is illustrated in Figure 1 and consists of a copper TSV with 5um diameter and 25um height and landing pads with 6um×10um and 1um thick. This test case constrains the current flow into the structure from the top-left corner and out of the bottom-right corner. A simplified grain structure is used for the TSV and nearby wire connections. This grain structure was chosen to approximate the grain structure described in [8].

Electromigration in a metal depends strongly on its grain structure, which in turn depends on the fabrication process and subsequent thermal and electrical history [8]. In our TSV model, we construct a simplified regular grain structure on the TSV and wires. The grains are 1.9um cubes separated by 0.1um-thick grain boundaries, both defined interior to the bounding surfaces of the TSV as illustrated in Figure 1. These dimension are consistent with the measured mean value of TSV grain size presented in [8]. In principle, the model could represent grains of any shape and size distribution. We believe this simplified structure is sufficiently detailed to reflect the essential basic physics intrinsic to the actual grain structure. Different activation energies are used for the grains and the grain boundaries. For grain boundaries, the activation energy, E_{A}(gb), a small value reflects high diffusion in grain boundaries; and for grains, the activation energy, E_{A}(g), a large value reflects low diffusion in grains. Since no specific activation energies have been reported for TSVs to date, we use values reported for wires: 0.9eV for E_{A}(gb) and 2.1eV for E_{A}(g) [12].

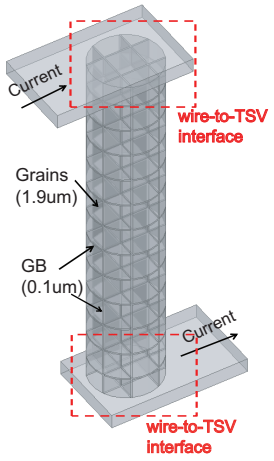


Fig. 1. Our TSV model with uniform grains and grain boundary (GB).

C. Stress Modeling

The EM atomic flux ($\mathbf{J_s}$ in Equation (6)) is affected by the gradient of hydrostatic stress. A TSV-based 3D connection can be influenced by the following three types of stress:

- Back flow stress: Back flow stress is generated by EM [14]. The atom flux leads to a distribution of atomic concentration along the 3D connection, where the atomic depletion generates tensile stress, and the atomic accumulation generates compression. This stress gradient tends to move the atoms back toward their original locations, thereby forming a "back flow" of atoms. This is the opposite to the atomic flux caused by the electric current density. Equation (7) is a linear equation which relates the back flow stress to the atomic concentration taken from the Korhonen model [15]. Our simulations described in Section IV-D show the significance of back flow stress, which also increases over time due to the EM-induced atomic concentration evolution.
- Thermal stress: Thermal stress is caused by the coefficients of thermal expansion (CTE) mismatch among different materials. Mechanical expansion can occur due to a temperature change, which may be caused by joule heating or the heat flux through the TSV. The motivation to study thermal expansion stress was driven by a concern that joule heating due to the large TSV current may cause self heating. In this work, we have ruled out self-heating as a significant effect in TSV modeling. Details will be discussed in Section IV-C.
- Residual stress: Residual stress is generated during the manufacturing process and "frozen" into the structure. For instance, thermo-mechanical stress [16] occurs during fabrication and thermal cycling of TSVs and is caused by the mismatch of thermal expansions coefficients among different materials. The residual stress can be released over time [17]. Thus, our EM modeling in this study ignores the residual stress.

D. Effective Resistance Modeling

One objective of modeling a single TSV structure in detail is to correlate simulations with experimental results. This is achieved if the modeling results can be compared with measurements from a TSV test site. There are typically many test sites which can be stressed in various ways and characterized using a simple 4-point probe to measure the effective resistance (Reff). To make such a comparison it is necessary to be able to calculate Reff from the simulation. We

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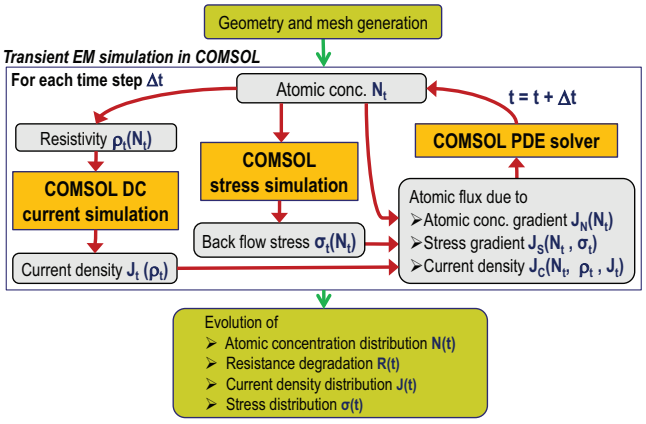


Fig. 2. Our transient EM simulation flow based on COMSOL Multi-physics toolset.

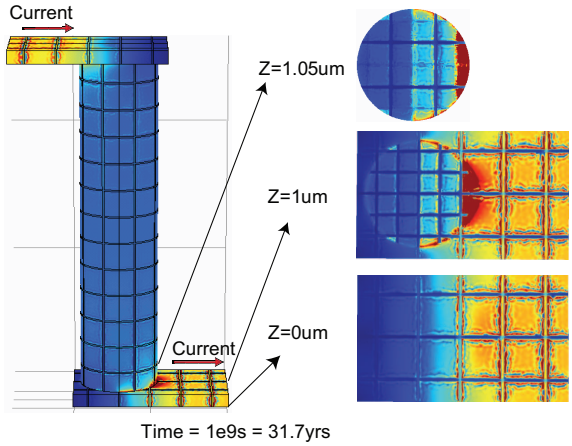


Fig. 3. Current density distribution in a TSV-wire structure, where TSV and wires contain grains. We show a 3D view and XY plane views.

can do this if we know the resistivity ρ as a function of x, y, z, and t.

In this work, the resistivity is modeled as a function of atomic concentration N. When the local atomic concentration decreases, the corresponding local resistivity increases; and likewise, when the local atomic concentration increases, the corresponding local resistivity decreases. Examples of resistivity function are depicted in Figure 9(a). The form and accuracy of our resistivity function need further investigation. Several variations of this function yield similar results, which will be discussed in Section V-B. We believe this function is a reasonable approximation to the relationship between the atomic concentration and resistivity. Regardless of the actual functional form, our modeling approach is able to easily accommodate any well-defined resistivity function.

III. Simulation Flow and Assumptions

We use COMSOL Multiphysics [11] for customized EM modeling and simulation, which enables us to combine user-specified EM PDEs, stress/resistivity modeling with multiple physics (e.g., electric current, joule heating, mechanical thermal expansion stress, etc). An overview of our transient EM simulation flow is illustrated in Figure 2. This flow starts by defining and creating the geometries of the TSV, connection wires, grains, and grain boundaries. This is followed by the mesh generation. EM transient simulation is then performed as an iteration loop shown with the red arrows in Figure 2. At each time step Δt, atomic concentration $\mathrm{N_t}$ at time t and the resistivity distribution $\rho_\mathrm{t}(\mathrm{N_t})$ are calculated based on the resistivity function. COMSOL is used to calculate the current density distribution $\mathrm{J_t}$ using the resistivity distribution ρ_t. In addition, COMSOL performs stress simulation based on user-specified equations. Next, the atomic fluxes are updated: (1) Flux $\mathrm{J_c}$ due to current density $\mathrm{J_t}$, atomic concentration $\mathrm{N_t}$, and resistivity ρ_t; (2) Flux $\mathrm{J_s}$ due to gradient of hydrostatic stress; and (3) Flux $\mathrm{J_N}$ due to atomic concentration gradient, which tends to balance $\mathrm{J_c}$. Finally, COMSOL solves the EM PDEs and calculates the atomic concentration at time $\mathrm{t} + \Delta\mathrm{t}$. In this transient analysis, COMSOL determines the iteration times and step for convergence automatically. Using this flow, we are able to derive the evolution of each physics phenomenon due to EM over time, e.g., atomic concentration gradient, effective resistance degradation, electric current, stress, etc.

For simplicity, a number of assumptions were made. We assume uniform grain and grain boundary geometry. We do not consider the grain orientation and grain/grain boundary propagation. Nucleation sites for void and hillock formation were not included. We assume that grain and grain boundaries have the same initial atomic concentration and resistance. Our simulation shows negligible thermal gradient from joule heating and thermal stress due to high thermal conductivity of copper TSVs. Quantum effects such as atomic tunneling through grain boundaries were ignored. In principle, all of these effects which were simplified or ignored can be included in the model later.

IV. Multiphysics Simulations

A. Simulation of Electrical Current

An existing study [1] has been conducted on detailed current density simulation of TSV-wire 3D connection and has shown a large amount of current concentration at the TSV-wire interfaces (called current crowding). However, this work did not consider the time-dependent atomic movement due to EM and grain structures, nor the resulting resistance degradation due to atomic depletion. Our EM modeling approach is able to simulate the time-dependent detailed current density distribution due to EM-induced atomic depletion/accumulation in grains and grain boundaries: the atomic depletion increases the electric resistance, which then reduces the local current density and slows down the atomic depletion. In addition, since grain boundaries have lower activation energy than grains, atoms flux is faster in grain boundaries than in grains. This results in higher resistance and thus lower current density in grain boundaries than in grains.

An example of how EM alters the current density is plotted in Figure 3. First, the bottom and top TSV-to-wire interfaces are subject to higher current density due to current crowding. In particular, current is significantly crowded at the bottom-right corner for Z=1um. Second, current concentrates in the grains rather than grain boundaries due to the resistivity distribution (high in grain boundaries and low in grains).

B. Simulation of Atomic Transport

Our EM modeling approach is able to simulate the time-dependent atomic accumulation and depletion due to multi-physics EM phenomenon. The atomic concentration at the top and bottom TSV-wire interfaces at time=1e5s and 1e7s are plotted in Figure 4. Here 60mA current enters at the top-left and exits from the bottom-right of the 3D

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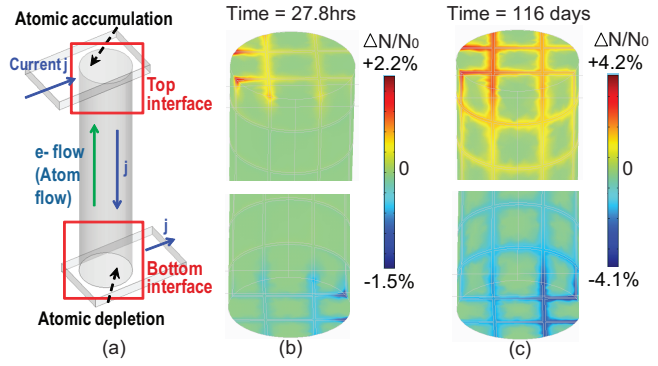


Fig. 4. Atomic concentration on TSV-based 3D connection. (a) electron/atom flow directions, (b) concentration at time=1e5s, (c) time=1e7s.

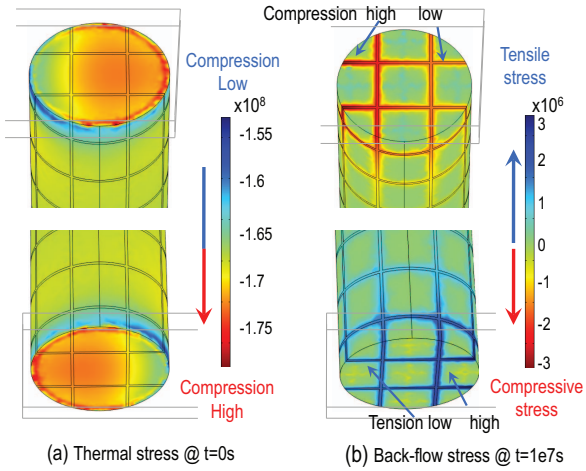


Fig. 5. Stress simulation results. (a) thermal expansion stress only at time=0s, where compressive stress gradient occurs at both interfaces. (b) back flow stress only at time=1e7s, where tensile stress occurs at the top and compressive at the bottom. Note that the tensile stress has positive value, and the compressive stress has negative value.

connection. The simulation results explicitly demonstrate the impact of electric current on atomic flux, in particular the current crowding effects. Since electrons flow from the bottom right to top left, copper atoms move from bottom right to top left due to momentum exchange after electron collision. Because of a large amount of current crowding at top-left and bottom-right TSV-wire interfaces [1], the bottom-right interface has depleted atoms ($\Delta N/N<0$), and the top-left interface has accumulated atoms ($\Delta N/N>0$). In addition, since atoms move faster along grain boundaries than grains, atomic depletion and accumulation occur along grain boundaries. Over a long period, more atoms will accumulate or deplete at the interfaces as shown in Figure 4(b).

C. Simulation of Thermal Expansion Stress

Due to the high thermal conductivity of copper, the thermal gradient is very small inside the copper TSV. To verify this, we simulate the joule heating in a TSV with 60mA current. The structure consists of three silicon layers (each is 25um thick), two inter-layer dielectric (ILD) layers (each is 4um thick), a TSV liner (SiO_2 with 0.2um thick), and a copper TSV with two landing wires. The heat sink is assigned at the top surface with the heat transfer coefficient of 25e3W/(m^2·K). Related experiments showed that a small thermal gradient is observed in the ILD layers, landing wires, and the TSV. Across this structure, the temperature varies from 349.84K to 349.90K. The resulting thermal gradient across the TSV between the top and bottom landing pads is only 0.04K.

The thermal expansion stress and its gradient impact on EM atomic flux are simulated for TSV-based 3D connection, where the copper TSV and wires are buried inside the silicon substrate, and the insulation layer (SiO_2) and barrier layer (TiN) surround the 3D connection. The thermal expansion stress is generated by joule heating effect, where the current flows from the top-left wire, through the TSV, and out of the bottom-right wire. In this case, the thermal expansion stress presents an initial compressive stress at the 3D connection. However, the stress gradient from thermal expansion primarily occurs at the TSV-wire interfaces, not inside the entire TSV connection. In addition, the stress gradient does not vary significantly over time. Therefore, thermal expansion stress presents minor impact on the overall EM atomic transport. The simulation result is plotted in Figures 5(a), where only the thermal expansion is considered. Note that in this stress distribution, the tensile stress has positive value, and the compressive stress has negative value.

We first observe that compressive stress on both TSV top and bottom is observed along the 3D connection initially. Since the

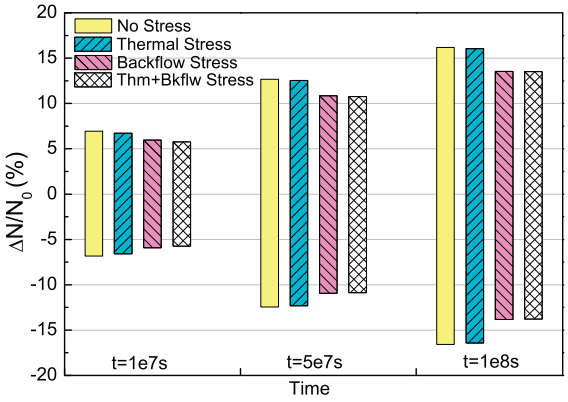


Fig. 6. Stress impact on atomic concentration evolution ($\Delta N/N_0$) at t=1e7s, 5e7s, and 1e8s. Back flow stress slows down the concentration degradation, but thermal stress has minor impact.

connecting wires are asymmetric on top and bottom, the top-right and bottom-left TSV-wire interfaces experience larger compressive stress (with short connecting wires) than the top-left and bottom-right interfaces (with long connection wires). In addition, due to the SiO_2 layer, the TSV-silicon interfaces experience less compressive stress on the top and bottom. Thermal expansion stress does not vary significantly over time. Second, we note that the stress gradient due to thermal expansion presents minor impact on atomic degradation. The stress impact on atomic concentration degradation ($\Delta N/N_0$) is plotted in Figure 6 at time=1e7s, 5e7s, and 1e8s. At each time, we compare the range of $\Delta N/N_0$ with (1) no stress, (2) thermal expansion stress only, (3) back flow stress only, and (4) both thermal expansion and back flow stress. From the comparison between "No Stress" and "Thermal Stress" bars, we note that $\Delta N/N_0$ does not change significantly by thermal stress.

D. Simulation of Back Flow Stress

The back flow stress and its gradient impact on EM atomic flux are simulated for TSV-based 3D connection, where we keep the same 3D connection structure as the thermal expansion stress, and apply the

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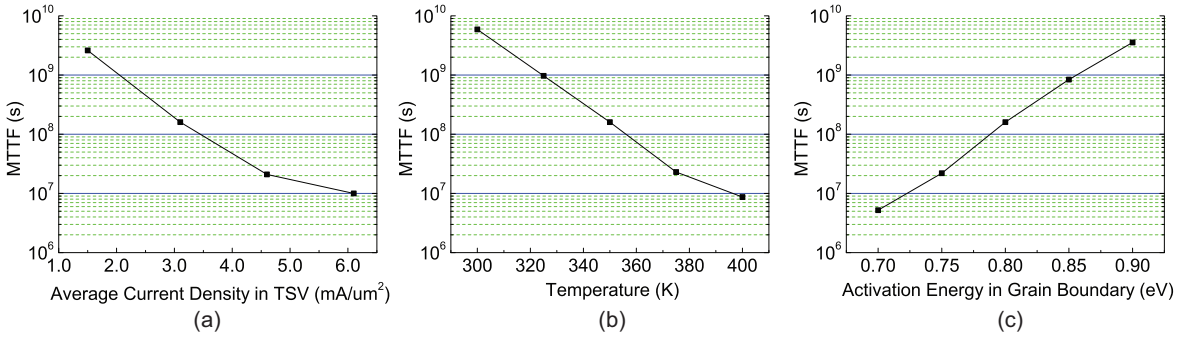


Fig. 8. (a) MTTF vs. average current density, T=350K. (b) MTTF vs. temperature, j_{TSV}=3.1mA/um^2, (c) MTTF vs. E_A(gb).

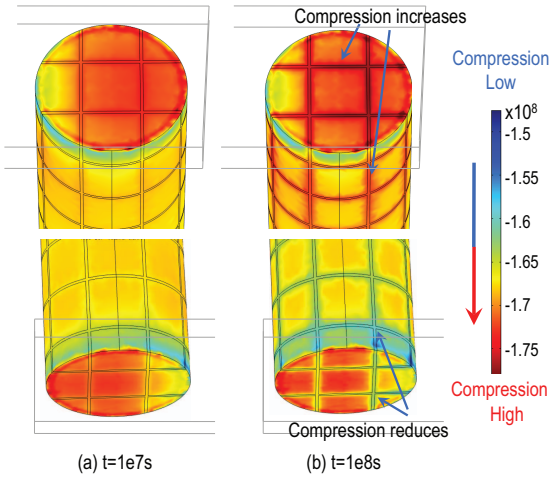


Fig. 7. Stress simulation with both thermal expansion and back flow stress considered. (a) at t=1e7s, (b) at t=1e8s. Back flow stress changes over time, resulting in compressive stress increased on the top and reduced on the bottom.

atomic flux equations due to back flow stress, current crowding, and concentration gradient. This back flow stress is originated from the compression at atomic depletion and tension at atomic accumulation, and presents long-time effect on atomic flux due to EM.

The simulation results are plotted in Figure 5(b) with t=1e7s. We first observe that compressive stress occurs at the top TSV-wire interface and tensile stress at the bottom TSV-wire interface. The back flow stress generates a considerable stress gradient through the entire TSV connection, especially along the grain boundaries where the atomic accumulation and depletion are likely to take place. Second, the TSV top-left and bottom-right corners suffer larger compression and tension than the top-right and bottom-left corners, respectively, which is mainly due to the current crowding impact on the atomic degradation. Third, from the comparison between "No Stress" and "Backflow Stress" bars, we note that $\Delta N/N_0$ changes significantly by back flow stress. This clearly shows that the atomic movement is reversed by the back flow stress.

We perform another EM simulation, where we consider both thermal expansion stress and back flow stress. The simulation results are plotted in Figure 7 at t=1e7s and t=1e8s. The thermal expansion stress creates compressive stress along the TSV initially. When EM effects last longer, however, back flow stress dominates the stress gradient, which in turn causes the top TSV-wire connection to be more compressive and the bottom connection less compressive. In addition, referring to Figure 6 and by comparing back flow stress only vs combined stress, we note that thermal expansion stress does not add any significant contribution. This confirms our observation that it is reasonable to ignore thermal expansion stress in EM simulation.

V. TSV Failure Analysis

In this section, we present failure analysis discussion that falls into two categories: one is the mean-time-to-failure (MTTF), which is defined as the 10% deviation of atomic concentration over the initial concentration N_0; the other is resistance degradation due to EM effects. Again, transient EM modeling is applied to the structure shown in Figure 1. More general TSV connection structure for full-chip power integrity analysis will be discussed in Section VI. The impact of various design and modeling factors on EM reliability will be investigated, including the current density, temperature, activation energy, resistivity function, and grain structure on the wire.

A. MTTF Analysis

The impact of TSV current density (j_{TSV}) on MTTF is shown in Figure 8(a), where j_{TSV} increases from 1.5mA/um^2 to 6mA/um^2, and temperature is 350K. As a result, the EM lifetime of the TSV dramatically reduces from 2.6e9s to 1.0e7s. This high current density accelerates the atomic depletion and accumulation and decreases the EM lifetime. For P/G TSVs, which can carry the current density larger than 5mA/um^2, the EM reliability may become critical.

Temperature also plays an important role in atomic concentration and EM reliability. Diffusivity D exponentially depends on temperature (Equation (8)). In addition, atomic fluxes are affected by temperature based on Equations (4) to (6) and thermal gradient in Equation (5). To analyze the impact of temperature on EM, the current value is kept constant, and the temperature is increased from 300K to 400K. This temperature range is affected by both the power density from neighboring devices and the joule heating of TSV. The impact of temperature on EM lifetime is shown in Figure 8(b). As the temperature increases from 300K to 400K, the MTTF dramatically reduces from 5.9e9s to 8.7e6s.

The diffusivity D exponentially depends on $-E_A$ (Equation (8)). Small activation energy of grain boundary (E_A(gb)) results in fast atomic diffusion, thus degrading the EM lifetime. The impact of E_A(gb) on MTTF is depicted in Figure 8(c), where E_A(gb) reduces from 0.9eV to 0.7eV. When E_A(gb) decreases, the MTTF dramatically reduces from 3.55e9s to 5.2e6s. This curve shows the exponential impact of activation energy on atomic flux.

B. Evolution of TSV Resistance

The resistivity evolution model deals with how TSV resistance degrades over time during EM lifetime. In our transient EM modeling

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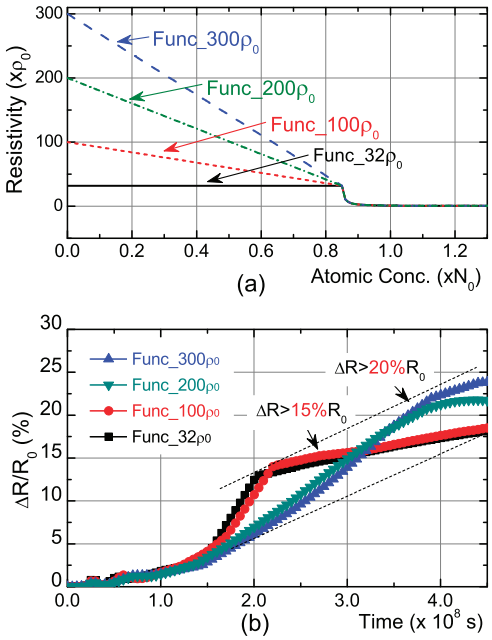


Fig. 9. Impact of resistivity function on resistance degradation. (a) four resistivity curves (Func_300ρ_0, Func_200ρ_0, Func_100ρ_0, and Func_32ρ_0), (b) corresponding resistance degradation.

approach, the resistivity is updated after each time step according to the local atomic concentration. These latest resistivity values change the current density distribution, which in turn changes the atomic transport and atomic concentration distribution. In this section we compare four possible resistivity evolution functions and show the impact on TSV resistance degradation. Most existing studies assume that a void occurs when the atomic concentration drops down to a specific percentage. However, there is no clear answer on how low the concentration threshold should be to declare a void.[1] Therefore, we describe a continuous resistivity increase when atomic concentration decreases, but vary the "infinite" resistivity values from $32\rho_0$ to $300\rho_0$ (see Func_32ρ_0 to Func_300ρ_0 in Figure 9(a)). The simulation results of $\Delta R/R_0$ shown in Figure 9(b) reveal a consistent resistance degradation over time among the 4 models (ΔR can reach to more than 20% R_0 due to EM). In addition, we observe that higher ρ_0 values lead to more resistance degradation. We note that the resistivity function with $32\rho_0$ shows reasonable accuracy and thus is used in this study.

C. Adding Grains in Wires

The TSV resistance evolution for a test case shown in Figure 1 that has no grains in the wires is plotted in Figure 10. This test case corresponds to the bamboo wire structure, where wires have no grains. We observe from Figure 10 (= red line) that at the early time period, TSV effective resistance increases rapidly. After 3e8 seconds, however, the resistance reaches to a saturation value. This is due to an equilibrium of atomic flux divergence between the opposing current stress and atomic concentration gradient. Up to 12% TSV resistance degradation is observed in this case.

[1]Our proposed modeling approach is able to integrate more complicated resistivity function if, in the future, more comprehensive study and measurement data on resistivity function of atomic concentration are available.

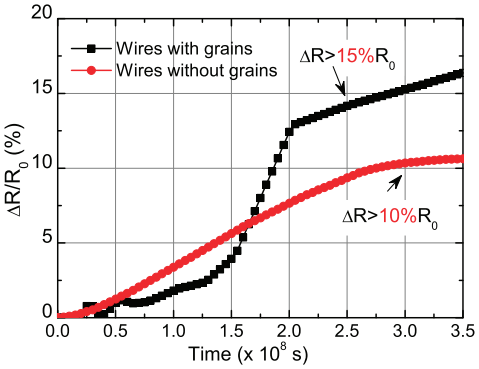


Fig. 10. Evolution of TSV effective resistance (RTSV) when wires have or do not have grains.

This bamboo structure is extended to include grains in the wires to investigate a non-bamboo wire structure. Wires and the TSV have similar grain structure. In the grained-wire case, atoms can now transport between wires and the TSV. Two curves of TSV resistance degradation are plotted in Figure 10 (= black line). When wires are non-bamboo (with grains), TSV resistance increases by more than 15% compared with the initial value. This larger TSV degradation is due to the high local current density in the wires when wires have grains. Whereas, for bamboo wires (no grains in wires), the TSV resistance degradation is approximately 10% due to the uniform current density in the wires.

VI. Long-term Reliability of Full-Chip 3D PDNs

In this section, a transient analysis methodology for chip-scale 3D PDN EM reliability is implemented.

A. Full-chip PDN Reliability Analysis Methodology

The analysis flow for our TSV-based 3D PDN long-term reliability simulation is depicted in Figure 11. The EM modeling approach discussed in Section II is applied to calculate TSV effective resistance. A library of TSV model which computes and returns the effective TSV resistance based on currents into and out of the TSV is generated. This methodology allows us to analyze the IR drop degradation of 3D PDNs. The entire flow consists of the following steps. First, an initial netlist for 3D PDN is generated from SPICE simulation. The initial TSV resistance is calculated using $\rho \times l/s$. Second, the initial current density of the PDN is calculated. Third, transient PDN reliability analysis is performed over multiple iterations. For each iteration at time t, SPICE simulation is performed using the updated TSV resistance value. The current density distribution and voltage drop are analyzed. Finally, the TSV resistance increase is calculated by using the TSV resistance library.

The simple TSV modeling structure shown in Figure 1 is extended to consider neighboring wires connecting to all four terminals of the TSV as well as the relative P/G C4 location to the TSV. The TSV resistance degradation due to EM is mainly affected by two factors: current density through the TSV, and the current distributed to neighboring wires, which dictates how the current is crowded at the TSV-to-wire interfaces. Two sample cases are shown in Figure 12. In Figure 12(a), current is provided from the C4 directly below the TSV and is distributed to the four neighboring wires. In Figure 12(b), TSV and C4 are offset, and current can either come into or flow out from the TSV neighboring wires.

We build and utilize a library to efficiently calculate the effective TSV resistance degradation for full-chip PDN analysis. The input

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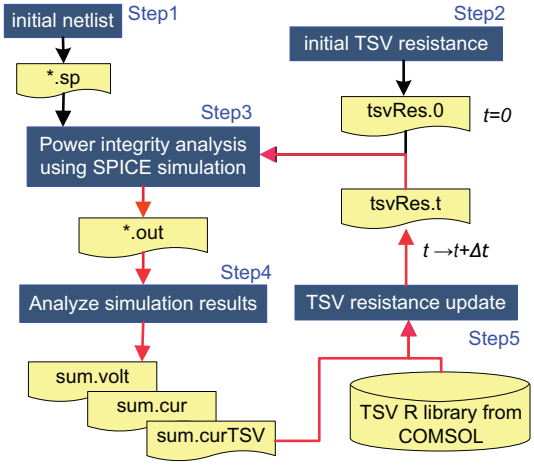


Fig. 11. Transient reliability analysis methodology for long-term power integrity of 3D PDNs.

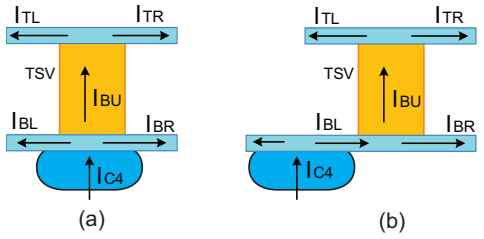


Fig. 12. Two sample cases for multi-terminal current flows in our P/G TSV structure.

to the library is the current density and direction at each corner connecting to the TSV, and the output is the TSV effective resistance curve with respect to the time. A large number of simulations are performed, where each case contains the combination of current density and direction for the wires (e.g. I_{BL}, I_{BR}, I_{TL}, I_{TR}, or I_{C4}). Curve fitting is applied on these data points to derive analytical equations for the TSV resistance. By querying the library with the in-situ TSV currents, the corresponding TSV model is chosen to calculate the effective resistance of the TSV at a given time and the currents through the neighboring wires and TSVs.

The TSV effective resistance degradation under the multi-terminal current flow is shown in Figure 13, where the TSV and the C4 are aligned, and temperature is set to 380K. To study the TSV current density effect, current densities in neighboring wires are kept balanced, where I_{BL}=I_{BR} and I_{TL}=I_{TR} in this experiment. The TSV resistance evolution curves are plotted in Figure 13 under different TSV current values. We observe that with the TSV current increasing from 96mA to 240mA, the TSV resistance degradation becomes much larger. In particular, the TSV resistance value increases from $33m\Omega$ to $53m\Omega$ with I_{BU} of 240mA.

B. Transient Power Integrity Analysis

Our 3D power delivery network is represented as a 3D resistive mesh. Each TSV is represented as a single resistor having an effective resistance Reff. Our time-dependent TSV library described earlier is able to capture the impact of both current crowding and EM on TSV resistance degradation over time. The power wire is 6um wide and 1um thick. The TSV has 5um diameter and 25um height. Grains in both TSVs and wires are 1.9um large with 0.1um grain boundary.

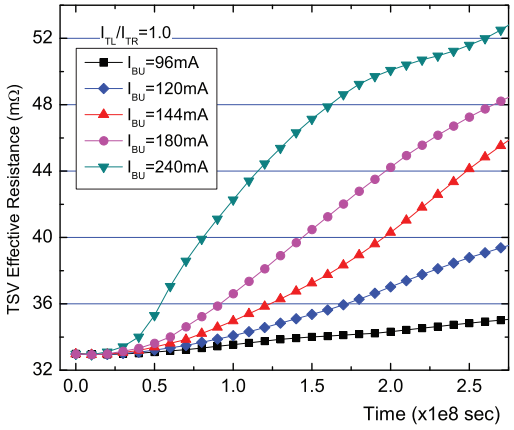


Fig. 13. Impact of TSV current density on EM in multi-terminal structure. TSV and C4 are aligned.

Temperature is set to 380K. Current density is obtained by SPICE simulation.

Our transient power integrity analysis is applied to a 3D PDN, which is 2-die stack, each die having 1.4mm x 1.4mm footprint area. The global PDN contains 16x16 power wires (yellow and blue lines in Figure 14) with 4x4 TSVs and C4s (white boxes). C4s provide 1.0V power supply to the bottom die (die-1), and P/G TSVs deliver the supply to the top die (die-2). IR drop maps for die-2 at t=0 and t=1e9s are shown in Figures 14(a) and 14(b), respectively. Due to the EM-induced TSV resistance degradation, IR drop increases over time. Especially at the current hot spot (top right corner of the design), the TSV resistance increases faster over time than the current cool spot. As a result, the maximum IR increases from 23.0mV to 43.2mV.

C. Large-scale 3D IC PDN Analysis

The simulation results of PDN long-term reliability on five large-scale 3D PDNs are shown in Table II. The maximum IR drop degradation in the top die includes the following results: (1) IR drop at t=0s without EM and current crowding (Column 8), and (2) IR drop due to EM and current crowding at times t=0s, 1e8s, 2e8s, and $>$ 10yrs (Columns 9-12). The percentage increase of IR drop is computed with respect to the IR drop with no current crowding at time t=0s.

We observe that IR drop degradation is affected by both current crowding and EM. First, current crowding at t=0s (Column 9) increases the maximum IR drop by 5.5% to 8.5% compared with the initial IR drop that does not include EM and crowding. Second, for each design after approximately 3 years (Column 10), maximum IR increases from 5.9% to 14.1%. This degradation continues increase over time. After approximately 10 years (Column 12), the PDNs suffer between 10.5% to 25.6% degradation due to IR drop. Finally, the bottom die also shows similar IR drop. The chip-scale PDN analysis requires tens of iterations for the entire PDN network with TSV resistance update for each iteration. The SPICE simulations are within a few seconds.

VII. Conclusions

In this paper, we studied electromigration (EM) reliability for TSV-based 3D ICs and 3D power delivery networks (PDNs). We proposed a transient EM modeling approach to simulate time-dependent atomic concentration and effective resistance degradation. This method iteratively updates the resistivity based on the latest

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TABLE II

EM RELIABILITY ANALYSIS OF LARGE-SCALE 3D PDNS INCLUDING THE FOOTPRINT, POWER DENSITY, DEGRADATION OF MAXIMUM IR DROP WITH PERCENTAGE INCREASE, AND RUNTIME FOR EACH SPICE SIMULATION.

Design	Footprint	Pwr dens. top	Pwr dens. bot	Power grid	#TSVs	#C4	IR_top in mV(% increase)					IR_bot	Runtime per itr.
							no crowding	crowding+EM					
	mm^2	W/mm^2		(#×#)			t=0	t=0	t=1e8s≈3.2yrs	t=2e8s≈6.3yrs	t >10yrs	mV	sec
PDN1	5×5	0.57	0.57	50×50	144	144	23.7	25.0 (5.5)	25.1 (5.9)	25.6 (8.0)	26.2 (10.5)	20.8	3
PDN2	6×6	0.80	0.75	60×60	225	225	23.1	24.6 (6.5)	24.9 (7.8)	25.8 (11.7)	26.7 (15.6)	16.2	6
PDN3	9×9	0.80	0.80	90×90	484	484	45.8	48.7 (6.3)	50.1 (9.4)	52.7 (15.1)	54.5 (19.0)	31.0	20
PDN4	11×11	0.71	0.91	110×110	729	729	29.3	30.9 (5.5)	31.2 (6.5)	32.1 (9.6)	32.9 (12.3)	32.9	41
PDN5	15×15	0.47	0.49	150×150	1369	1369	57.5	62.4 (8.5)	65.6 (14.1)	69.6 (21.0)	72.2 (25.6)	44.7	64

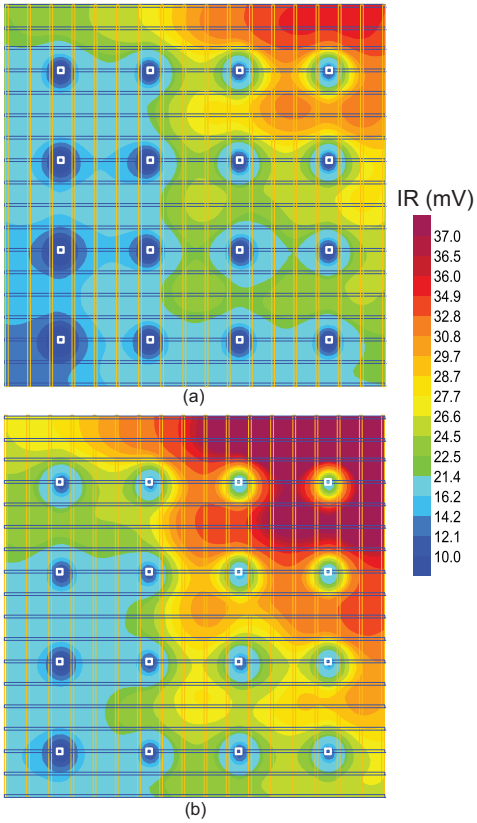


Fig. 14. EM-induced IR drop degradation on die-2 for a 3D PDN. (a) t=0 (max noise: 23.0mV) and (c) t=1e9s (max noise: 43.2mV).

atomic concentration, simulates the electric current, stress gradient, and calculates the atomic concentration for the next time step. Grains and grain boundaries were also included in the TSVs and neighboring wires. In addition, we developed a transient power integrity analysis flow for chip-scale 3D PDNs lifetime prediction, which integrated our EM modeling approach, simulated the IR-drop evolution, and analyzed the maximum IR degradation due to current crowding and EM.

Our observations are as follows: (1) Atomic depletion or accumulation occurs at wire-to-TSV interfaces, in regions of high current density and in particular where current crowding occurs; (2) Back flow stress gradient presents non-negligible impact on atomic flux, whereas the thermal expansion stress has minor impact; (3) High temperature, large current density, small grain size, or low activation energy of grain boundaries can accelerate EM, thus shortening the lifetime of TSVs; (4) EM-induced degradation of TSV effective resistance is simulated, and the neighboring wires with grains accelerate the EM degradation; (5) For chip-scale PDN reliability analysis, under some typical conditions, current crowding and current-driven EM can increase the maximum IR drop significantly over time. The modeling and PDN reliability analysis flow described in this paper can also be used for other 3D structures subject to high current density and high thermal and mechanical stress.

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